CP001824.1:1117181-1133702
Sphaerobacter thermophilus DSM 20745

B12-binding Radical_SAM Methyltransf_11 MTS Methyltransf_25 B12-binding Radical_SAM B12-binding Radical_SAM ? ? ParE_toxin PhdYeFM_antitox NTP_transf_2 ABC_membrane ABC_tran 2-Hacid_dh_C Beta-lactamase2 Na_H_antiporter TENA_THI-4

AP018207.1:1097423-1111939
Calothrix brevissima NIES-22

? HTH_31 MoeA_N MoCF_biosynth MoeA_C ? RDD Ribosomal_S18 RNB HTH_RNase_II wH_RNase_II CSD_RNase_II ParE_toxin ? Glycos_transf_2 Glyco_trans_2_3 ECH_1 ? DciA ? PspA

AP018223.1:1024859-1037851
Nostoc linckia NIES-25

Acetylase Orn_Arg_deC_N DAC Prenyltransf ParD_antitoxin DUF3143 DnaJ ParE_toxin RelB-like DnaJ DnaJ_C HSP70 Peptidase_M28 Peptidase_M28 ? ? DUF559

CADCTY010000587.1:899-893
uncultured Leptolyngbya sp.

? ParE_toxin

BEHR01000009.1:71700-83957
bacterium HR15

? Lactamase_B RNase_J_b_CASP RMMBL RNase_J_C Clp_N Clp_N AAA AAA_lid_9 AAA_2 ClpB_D2-small ? ? ? RelB-like ParE_toxin ? ? ? ? ?

DMPE01000108.1:1-4473
Planktothrix sp. UBA8407

Rsd_AlgQ Helicase_C ? ParE_toxin ? HSP33

DNKF01000157.1:4217-4045
Methylococcaceae bacterium

META META ParE_toxin DUF6290 Aminotran_1_2 Aminotran_1_2 UvrB_inter ResIII

DOWK01000431.1:147-7027
Blastocatellia bacterium

Acetyltransf_11 Hexapep Hexapep Hexapep Hexapep ParE_toxin RelB-like FAD_binding_4 FAD-oxidase_C PIN PhdYeFM_antitox CHRD FG-GAP_3 FG-GAP_3 FG-GAP_3 SDF Uma2

DQWU01000172.1:2537-3341
Candidatus Bathyarchaeota archaeon

AAA DSHCT ParE_toxin

DQZL01000585.1:252-2238
Thioploca sp.

ANG-1 CTHRC1_C ParE_toxin DUF2281

DASSAY010000031.1:124-3943
Balneolaceae bacterium

ParE_toxin PhdYeFM_antitox Aminotran_3 Acetyltransf_7 Ala_racemase_N

DASUBQ010000149.1:2006-9452
Terriglobales bacterium

Glyoxalase LplA-B_cat OmpA ? DinB E1_dh ? ParE_toxin PhdYeFM_antitox Transketolase_C Transket_pyr DUF2007 Cytochrome_CBB3

DASTWU010000182.1:52-12232
Gemmatimonadaceae bacterium

? HTH_8 AAA_lid_14 NtrC-AAA Methyltransf_25 Acetylase PhdYeFM_antitox ParE_toxin PadR MacB_PCD FtsX MacB_PCD FtsX FR47 DUF4126 DUF2911 OGG ?

DASZBR010000005.1:86063-98211
Gammaproteobacteria bacterium

TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C YecM Pyr_redox_2 HTH_1 LysR_substrate HTH_18 GyrI-like Beta-lactamase RelB-like ParE_toxin ? MFS_1 Dabb HTH_1 LysR_substrate Acetylase Acetylase Fic DUF4172

DASZFI010000197.1:3741-6146
Gemmataceae bacterium

Proton_antipo_N Oxidored_q2 Oxidored_q3 NADHdh SMC_ScpB ParE_toxin PhdYeFM_antitox NTP_transferase LbH_GLGC

DASZIA010000098.1:1-7723
Planctomycetaceae bacterium

? PQQ_2 PQQ_2 PhdYeFM_antitox ParE_toxin FloA ? Glyco_transf_4 Glycos_transf_1 MoCF_biosynth CinA_KH CinA

DATAHG010000705.1:1-3550
Thermoanaerobaculia bacterium

DEAD Helicase_C WH_Lhr DEAD_assoc ParE_toxin RelB-like dUTPase ?

DTUX01000200.1:3433-5579
Aquificales bacterium

N6_Mtase HsdM_N HepT-like Polbeta Methylase_S ParE_toxin RHH_1 Unstab_antitox SWI2_SNF2 REase

DYDP01000109.1:150-1516
Rickettsia endosymbiont of Bembidion lapponicum

ParE_toxin PhdYeFM_antitox ?

DATEXL010000273.1:1706-8609
Polyangia bacterium

PhdYeFM_antitox ParE_toxin Kelch_4 Kelch_KLHDC2_KLHL20_DRC7 Kelch_4 Lipocalin_2 Methyltransf_23 Glyco_trans_1_2 GH97_N Glyco_hydro_97 GH97_C

DATFWS010000259.1:560-4222
Candidatus Binatia bacterium

? Rap1a SWIM PhdYeFM_antitox ParE_toxin Peptidase_S74 ABC_tran_CTD ?

DATDOF010000804.1:310-2816

Chitinophagaceae bacterium

HTH_18 AraC_binding ParE_toxin RelB-like HTH_18

DATGLZ010000295.1:1-5113
Chloroflexota bacterium

BPD_transp_1 BPD_transp_1 PhdYeFM_antitox ParE_toxin Sulfatase Glyco_hydro_39 LacI Peripla_BP_3

DATHAT010000173.1:1-3147
Tepidisphaeraceae bacterium

HATPase_c_3 ? ParE_toxin ? ? Peptidase_M29

DATGVO010000021.1:1-3797
Planctomycetaceae bacterium

Receiver Receiver His-Kinase His-Kinase PAS PAS Receiver ? PhdYeFM_antitox ParE_toxin

DAOLFT010000003.1:40550-53100
Candidatus Saccharibacteria bacterium

Mur_ligase_M ? Aminotran_5 Dala_Dala_lig_N Dala_Dala_lig_C Mur_ligase_M Mur_ligase_C Abhydrolase_1 DNA_binding_1 ParE_toxin RHH_1 tRNA_SAD tRNA-synt_2b HGTP_anticodon ChlI AAA_25 Zn_ribbon_LapB TRAM PIN DUF1232 UDG TauE

DATIGG010000362.1:400-4082
Thermodesulfobacteriota bacterium

HicA_toxin HicB_lk_antitox ParE_toxin PhdYeFM_antitox PhdYeFM_antitox HTH_3 PIN RHH_1 PIN RHH_1

DATHQB010000142.1:1-6381
Candidatus Sulfotelmatobacter sp.

? ? Acyl-CoA_dh_N Acyl-CoA_dh_2 PhdYeFM_antitox ParE_toxin ? Sortilin-Vps10 Sortilin-Vps10 Peptidase_S49

DATHMO010000257.1:7263-15967
Bryobacteraceae bacterium

DNA_pol3_delta_C DNA_pol3_delta LptE Semialdhyde_dhC Semialdhyde_dh CDP-OH_P_transf PS_Dcarbxylase MotA_ExbB RelB-like ParE_toxin DUF3738 Glyoxalase_7 ? Glucosaminidase ? HTH_8 AAA_lid_14

DATKAE010000023.1:3560-5008
Thermoanaerobaculia bacterium

STY-kinase PD40 PhdYeFM_antitox ParE_toxin TPRs

DATJXG010000114.1:1-6339
Terriglobales bacterium

Calcineurin ParE_toxin PhdYeFM_antitox Transketolase_C Transket_pyr Cytochrome_CBB3 Trans_reg_C Receiver His-Kinase His-Kinase HAMP ?

DATKTZ010000021.1:60502-67965
Acidimicrobiales bacterium

? HTH_30 GGDEF_2 ? DUF3024 UvrD-helicase Zn_ribbon_IS1595 DDE_Tnp_IS1595 Phage_int_SAM_3 Phage_integrase ParE_toxin PhdYeFM_antitox

DATLTK010000869.1:1-2775
Chloroflexota bacterium

? ParE_toxin RelB-like Gp53-like_C SH3_3

DATKJU010000297.1:106-4535
Thermodesulfobacteriota bacterium

Fer2_3 Fer2_2 ParE_toxin PhdYeFM_antitox Guanylate_cyc TPRs TPRs LemA

DASZRN010000071.1:6431-20347
Polyangia bacterium

Sigma70_r2 Sigma70_r1_2 ? ? CN_hydrolase LNT_N PNPOx_N UPF0020 THUMP RlmL_1st Unstab_antitox ParE_toxin ATP-synt_DE_N ATP-synt_VA_C ATP-synt_ab ATP-synt_ab_N ATP-synt ATP-synt_ab_C ATP-synt_ab ATP-synt_ab_N OSCP ATP-synt_B ATP-synt_B

DASZPW010000203.1:299-3527
Chitinophagaceae bacterium

RelB-like ParE_toxin Peptidase_S37 Pyr_redox_2

DASZLR010000113.1:489-8142
Pseudonocardiaceae bacterium

PhdYeFM_antitox ParE_toxin PhdYeFM_antitox His-Kinase His-Kinase Trans_reg_C Receiver Ppx-GppA ? AP_endonuc_2 Pro_dh

DATMDN010001211.1:89-2627

Polyangiaceae bacterium

ParE_toxin PhdYeFM_antitox Sigma70_r2 Sigma70_r4_2 ? ?

DATMQT010000087.1:28512-44300
Candidatus Dormiibacterota bacterium

Beta_helix ? ? ? Biotin_lipoyl E3_binding 2-oxoacid_dh Cytochrome_CBB3 Transket_pyr Transketolase_C ParE_toxin ? E1_dh CHB_HEX_C_1 Glyco_hydro_30 Glyco_hydro_30C CBM_6 LplA-B_cat Pyr_redox_2 Pyr_redox_dim S1 S1 S1 S1 S1 S1 PAP2

DATNBP010000178.1:10387-29882
Puia sp.

ORF6N ? ? ? ? DUF6443 ? ParE_toxin RelB-like HTH_3 Toprim_2 DNAG_N Zn_ribbon_DnaG HTH_Tnp_1 rve_3 rve HTH_21 Phage_int_SAM_1 IstB_IS21

DATFFW010000505.1:1-744
Candidatus Paceibacterota bacterium

PhdYeFM_antitox ParE_toxin

DATNHJ010000131.1:157-9605
Candidatus Acidoferrum sp.

Dehyd-heme_bind Peptidase_S10 Glyco_hydro_92N Glyco_hydro_92 RelB-like ParE_toxin BcrAD_BadFG GKRP_SIS_N GKRP-like_C ? PIN DivIC RRF AA_kinase ?

DATNEW010000020.1:53208-66841
Gemmatimonadaceae bacterium

? HhH-GPD DUF6496 PAP2 BTAD GerE ? ParE_toxin PhdYeFM_antitox ? ? MutS_V PAP2 FUSC-like FUSC_2 HlyD_D23

DASPRL010000041.1:1-6582
Terriglobales bacterium

Pyr_redox_dim ? HicA_toxin LplA-B_cat E1_dh HTH_31 ? ParE_toxin PhdYeFM_antitox Transketolase_C Transket_pyr Cytochrome_CBB3 ?

DASQFH010000286.1:1-6067
Vicinamibacteria bacterium

GGDEF ParE_toxin PhdYeFM_antitox GatB_N Asparaginase_C Asparaginase GatD_N PemK_toxin RHH_1 ?

DASQOW010000199.1:1-6150
Puia sp.

Ketoacyl-synt_C ketoacyl-synt PP-binding ACPS BD-FAE RuvC LPG_synthase_TM DUF3108 ParE_toxin RelB-like

DASRKP010000486.1:92-9200
Candidatus Limnocylindrales bacterium

OTCace_N OTCace Pribosyltran DUF4032 DUF5615 DUF433 N_methyl SBP_bac_10 ParE_toxin RelB-like FeS_assembly_P ParA Glyco_hydro_43 BNR_2 ? ? ?

DASRBR010000691.1:1-7214
Verrucomicrobiia bacterium

UDPGP Transp_cyt_pur ParE_toxin ? Amidohydro_1 Aldedh 23S_rRNA_IVP DHO_dh

JQIP01000038.1:121-9807
Sulfurovum sp. AS07-7

? Ribonuc_L-PSP Ribosomal_L21p Ribosomal_L27 ParE_toxin DUF6290 MMR_HSR1 GTP1_OBG AA_kinase DUF333 HTH_1 LysR_substrate N_NLPC_P60 SH3_6 NLPC_P60 HI0933_like

JACCWM010000100.1:1-14782
Lujinxingiaceae bacterium

HATPase_c_3 ? ? Transp_cyt_pur PhdYeFM_antitox ParE_toxin ? Amidohydro_1 Aldedh Aminotran_3 PseudoU_synth_2 S4 AbiEii Rv0078B

JACDGZ010000537.1:1-2243
Nitrospirales bacterium

Trypsin BPD_transp_2 ParE_toxin RelB-like

PNOL01000078.1:1-5034
Anaerolinea sp.

HD ParE_toxin RelB-like DUF218 BPD_transp_1 BPD_transp_1 ABC_tran

JACONN010000005.1:776667-794328
Hormoscilla sp. GM7CHS1pb

? ? Uma2 Uma2 ? DUF2281 PhdYeFM_antitox ParE_toxin PhdYeFM_antitox ? Uma2 ? Acetylase 5_nucleotid_C Calcineurin

JACMLQ010000684.1:580-3661
Anaerolineae bacterium

CopG_antitoxin HTH_3 UPF0150 HicB ParE_toxin PhdYeFM_antitox Mrr_cat Methylase_S

JACMIR010000868.1:1-3776
Fibrella sp.

OEP HlyD_D23 dCMP_cyt_deam_1 ParE_toxin RelB-like RNB S1

JACNKN010000066.1:73214-92259
Candidatus Neomarinimicrobiota bacterium

Bgal_small_N LacZ_4 Glyco_hydro_2_C Glyco_hydro_2 Glyco_hydro_2_N Aminotran_3 APH Amino_oxidase Amino_oxidase Peptidase_M24 AMP_N ? DUF6290 ParE_toxin PAP2 Orn_Arg_deC_N Glyco_hydro_18 Ig_J_second Por_Secre_tail NamZ_N NamZ_C Peptidase_S13 DUF559 Lactamase_B_2

JACNJH010000257.1:88-11451
Candidatus Desulfatibia profunda

TPRs TPRs TPRs TPRs RNaseH-CATRA+Caspase Cytidylate_kin2 BON FecR Ppnp Flavin_Reduct LysM LysM SLT PhdYeFM_antitox ParE_toxin

JACVRQ010000519.1:1-3540
Coleofasciculus sp. Co-bin14

Glycos_transf_1 FTR1 ParE_toxin Uma2

JACJNX010000061.1:555159-566028
Cyanobacteria bacterium FACHB-63

GatC Transthyretin Lipase_GDSL_2 GHL10 FAD_syn Flavokinase AAA_3 AAA_lid_2 DUF2281 ParE_toxin ? SQS_PSY EVE GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C BipA_C

JACJQI010000052.1:37414-57137
Aulosira sp. FACHB-113

AAA_lid_3 AAA SH3_3 PG_binding_1 PG_binding_1 PG_binding_1 PG_binding_1 PG_binding_1 GAF PAS GGDEF EAL Transglut_core2 TPRs ATP-synt_VA_C ATP-synt_ab ATP-synt_ab_N ATP-synt_DE ATP-synt_DE_N ParE_toxin ? ? ? DO-GTPase2 Dynamin_N Band-7

JACSRX010000035.1:70973-82313
Chthoniobacterales bacterium

? ? ? ? ? ? RelB-like ParE_toxin ? ? ? ? ? ? ?

JABTUN010000001.1:16148-31667
Verrucomicrobiaceae bacterium

MFS_1 Glyco_hydro_77 YaeQ CHAD Asp_Glu_race OTCace_N OTCace Aminotran_3 ParE_toxin ? AA_kinase ArgJ Semialdhyde_dh Semialdhyde_dhC_1 ClpB_D2-small AAA_2 ? RecR_C Toprim_4 RecR_HhH

JADGCT010000232.1:1-3804
Desulfamplus sp.

? PKD K319L-like_PKD Big RelB-like ParE_toxin Bac_export_2

JADGCQ010000047.1:6428-23412
Magnetococcales bacterium

? ? Hpr_kinase_C Sulfotransfer_3 TPR_CcmH_CycH TPRs NTP_transf_5 Asn_synthase GATase_7 AAA ParE_toxin ? VTT_dom FeoB_associated NusB RsmF-B_ferredox Methyltr_RsmB-F MMR_HSR1 Dus His-Kinase His-Kinase PAS HTH_8 AAA_lid_14 NtrC-AAA Receiver

JABZMD010000043.1:1-4541
Thermobifida sp.

? ParE_toxin PhdYeFM_antitox ? Methyltransf_1N DNA_binding_1 Protoglobin Na_H_Exchanger

JACSDG010000229.1:140-2650
Magnetococcales bacterium

Phage_integrase Arm-DNA-bind_3 Phage_int_SAM_3 Phage_integrase RHH_1 ParE_toxin

JACOTU010000879.1:1-2640
Candidatus Poribacteria bacterium

Thymidylate_kin DNA_pol3_delta2 DNA_pol3_delta2 DNA_pol3_delta2 DNApol3-delta_C PhdYeFM_antitox ParE_toxin tRNA-synt_1g

JACOXK010000068.1:28-10828
Deltaproteobacteria bacterium

GIY-YIG ? Sigma70_r4_2 Sigma70_r2 ACR_tran ACR_tran HlyD_D23 HlyD_D23 HMBD OEP OEP PhdYeFM_antitox ParE_toxin ? Haem_bd Cu-oxidase_2 Cu-oxidase Cu-oxidase_3

JACPFC010000066.1:16559-37089
Deltaproteobacteria bacterium

DUF2283 AAA_14 DUF4143 ? ? ? ? DUF6290 ParE_toxin DUF4143 AAA_14 NAGPA ATP-synt_ab Rho_RNA_bind Rho_N Lyase_1 ASL_C2 Orn_Arg_deC_N DAP_epimerase DAP_epimerase DHDPS

JACPFL010000004.1:11857-23557
Deltaproteobacteria bacterium

LigD_Prim-Pol DUF72 GSDH UbiD_N UbiD UbiD_C UbiD_N UbiD UbiD_C NMT1 Amidohydro_3 ParE_toxin PhdYeFM_antitox MFS_1 ABC_tran ABC_tran ?

JACPPF010000034.1:16369-35069
Planctomycetota bacterium

GFO_IDH_MocA GFO_IDH_MocA_C3 Bact_transglu_N Transglutaminase 3keto-disac_hyd TPRs ? FGGY_N FGGY_C PhdYeFM_antitox ParE_toxin ? DUF4058 Secretin TPRs ? TRAM PIN HD_assoc HD HIT

JACPPF010000114.1:91950-113027
Planctomycetota bacterium

? DUF4175 DUF4175 SQHop_cyclase_N SQHop_cyclase_C Peptidase_S41 PDZ_Tricorn Tricorn_C1 Beta-prop_THOC3 PD40 PD40 ? Uma2 Aldedh ParE_toxin RelB-like DUF4058 Methyltransf_31 AMP-binding DUF6036 ? PQQ_2 PQQ_2

JACPSE010000154.1:210-10653
Deltaproteobacteria bacterium

DUF2442 DUF4160 IspD CarD_TRCF_RID CarD_C Acetylase Fer4 Int_N Phage_integrase ParE_toxin PhdYeFM_antitox HTH_17 ? Zn_ribbon_DnaG HTH_17 ? HTH_17

JACPTQ010000044.1:97430-115332
Deltaproteobacteria bacterium

HTH_3 DUF4011 AAA_11 AAA_11 AAA_12 MTES_1575 REase SWI2_SNF2 T1R_D3-like T1RH-like_C DUF3644 Methylase_S HsdM_N N6_Mtase HTH_31 ParE_toxin PhdYeFM_antitox HTH_11 WYL WCX FHA Prok-RING_1 ? ? ?

JACPVI010000057.1:37478-59540
Myxococcales bacterium

DUF1638 GAF_2 His-Kinase His-Kinase OmpA CarboxypepD_reg CarboxypepD_reg ? Transglutaminase FHA NtrC-AAA AAA_lid_14 HTH_8 DDE_Tnp_1_4 DDE_Tnp_1_4 PhdYeFM_antitox ParE_toxin GxGYxYP_C GxGYxYP_N_3rd GxGYxYP_N_2nd GxGYxYP_N AAA_16 STY-kinase ? OPT STY-kinase DUF3060 DUF3060 ADH_N ADH_zinc_N

JACPYD010000171.1:10602-17138
Candidatus Rokuibacteriota bacterium

NDK NMT1_2 Biotin_carb_N CPSase_L_D2 Biotin_carb_C BT_MCC_alpha Biotin_lipoyl ? HTH_3 PhdYeFM_antitox ParE_toxin ZnR

JACQED010000169.1:1696-8242
Chloroflexi bacterium

? HicA_toxin ATP-synt_DE ATP-synt_DE_N MreB_Mbl Put_DNA-bind_N CoA_binding HD_5 GAF_2 DUF1967 MMR_HSR1 GTP1_OBG ParE_toxin ? CTP_transf_like ?

JACQTE010000109.1:1-11566
Acidobacteria bacterium

PALP PRAI IGPS Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 ParE_toxin PhdYeFM_antitox Lactamase_B_3 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 ? TIG CarboxypepD_reg Peptidase_M10 HhH-GPD STY-kinase

JACQWA010000356.1:7007-6636
Candidatus Rokuibacteriota bacterium

LysM Peptidase_M23 Sigma70_r1_2 Sigma70_r2 Sigma70_r3 Sigma70_r4 PhdYeFM_antitox ParE_toxin Bac_luciferase Isochorismatase ECH_1 ECH_1 ?

JACRFD010000010.1:7316-26620
Ignavibacteriales bacterium

B12-binding Radical_SAM DUF4070 MvaI_BcnI N6_N4_Mtase N6_N4_Mtase MFS_1 MFS_1 CoA_binding_2 Succ_CoA_lig Ligase_CoA_2 ATP-grasp_5 Acetylase GatD_N Asparaginase Asparaginase_C GatB_N GatB_Yqey ParE_toxin RelB-like Sortilin-Vps10 Por_Secre_tail Lyase_aromatic YHS HMA Hydrolase E1-E2_ATPase HMA Memo

JACRCX010000047.1:114898-127017
Deltaproteobacteria bacterium

? ? ? MIF Amidohydro_3 Amidohydro_3 ? ? ParE_toxin PhdYeFM_antitox Aminotran_5 ACT PGDH_inter 2-Hacid_dh_C Radical_SAM

JACRDC010000044.1:25119-37943
Desulfarculus sp.

ParB_N PAPS_reduct 2-thiour_desulf HTH_ParB ParB_N Zn_ribbon_TFIIB ? GpA_ATPase GpA_nuclease PhdYeFM_antitox ParE_toxin ? Phage_portal_2 Phage_capsid_4 Phage_cement_2 Minor_tail_Z Phage_attach ?

JACRCO010000065.1:749-11938
Deltaproteobacteria bacterium

AbiEi_4 TfoX_N AHSA1 Aldo_ket_red Pro_dh PutA_N ORF6N PhdYeFM_antitox ParE_toxin DUF748 DUF748 Vgb_lyase Vgb_lyase DLH ? Glyoxalase_6 adh_short_C2 Isochorismatase

WGLZ01000095.1:2642-10651

Cyanobacteria bacterium RI_101

STY-kinase SSB PP2C_2 Copine VWA_2 ? NBD_C SBD_N DUF928 ParE_toxin ?

JADJOF010000005.1:230988-242291
Bdellovibrionales bacterium

P-loop_TraG CagE_TrbE_VirB ? ? ? ? ? TrbI ParE_toxin RH3_dom ? HTH_17 Phage_integrase ? TrwB_AAD_bind ?

JADJOJ010000010.1:653161-667712
Candidatus Leptofilum gracile

COX2 UbiA COX1 Cytochrom_C Cytochrome_B Fer4_7 FlpD NUDIX PhdYeFM_antitox ParE_toxin DUF1638 S-methyl_trans DinB_2 UPF0182 Cys_Met_Meta_PP PNPOx_N MazE_antitoxin

JADJPM010000004.1:3000212-3015960
Verrucomicrobiaceae bacterium

GFO_IDH_MocA GFO_IDH_MocA_C2 LON_substr_bdg AP_endonuc_2 Fer4_13 bPH_2 DUF218 SGL ParE_toxin ? Esterase Methyltr_RsmB-F RsmF-B_ferredox NusB VWA_2 BatA PSD1 PSCyt2 PSCyt1 HXXSHH CMAS

JADJTY010000001.1:44953-62092
Chitinophagaceae bacterium

Methyltransf_4 SDH_beta SDH_alpha DPPIV_N Peptidase_S9 Glyco_hyd_65N_2 Glyco_hydro_95_cat Glyco_hydro_95_C ? N-OB tRNA-synt_2 ? ParE_toxin ATP-synt_ab Rho_RNA_bind Rho_N ? cobW Rhodanese ? AIRS_C AIRS ? ?

JADJYZ010000006.1:195671-210610
Deltaproteobacteria bacterium

? ? WCX WYL HTH_11 ? ? DUF2237 Alpha-amylase Malt_amylase_C ParE_toxin ? Peptidase_S51 ACP_syn_III ACP_syn_III_C Chromate_transp Chromate_transp Cytochrom_C HYDIN_VesB_CFA65-like_Ig PEGA PEGA

JADHXM010000005.1:32186-40806
Desulfobacterales bacterium

Radical_SAM BATS ThiG ThiF ketoacyl-synt Ketoacyl-synt_C Receiver STY-kinase Usp ParE_toxin Antitox_RHH

JADHXM010000015.1:16800-33974
Desulfobacterales bacterium

PD40 PD40 TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs RNaseH-CATRA+Caspase Cytidylate_kin2 BON FecR Ppnp Flavin_Reduct LysM LysM SLT PhdYeFM_antitox ParE_toxin ? NucS_C Bac_DnaA_C CW_7

JAEUWE010000360.1:18-1941
Bdellovibrionaceae bacterium

TrbI ParE_toxin RHH_10 ? HTH_17

JAEUVS010000044.1:128-8105
Chitinophagaceae bacterium

? ? ? RelB-like ParE_toxin DUF1508 Aminotran_3 Hexapep Hexapep ? DUF3630 RsgA_GTPase RsgA_N VWA BatA

JAEUUC010000007.1:27511-38650
Cyclobacteriaceae bacterium

? ? ? DUF5684 Lycopene_cyc Lycopene_cyc Cyt_bd_oxida_II Polyketide_cyc2 ParE_toxin RelB-like DNA_photolyase FAD_binding_7 DUF6435 NUDIX SQS_PSY Amino_oxidase Sigma70_r2 Sigma70_r4_2

JAEURC010000080.1:645-14758
Anaerolineales bacterium

Peptidase_M24 Creatinase_N adh_short Lon_C Lon_lid AAA LON_substr_bdg Mur_ligase_C Mur_ligase_M ? ? PhdYeFM_antitox ParE_toxin DUF503 adh_short_C2 ? MurJ ? CopD TRAM_2 Radical_SAM UPF0004

JAEUQV010000086.1:1-15525
Candidatus Saccharibacteria bacterium

LPG_synthase_TM UbiA COX4_pro COX3 COX1 COX2_TM COX2 COX_ARM ParE_toxin DUF6290 FAD_binding_4 FAD-oxidase_C Pyr_redox_2 PaaX_M PDDEXK_1 UvrD_C UvrD_C UvrD-helicase MIP CDCA

JAFGJE010000084.1:7343-15566
Phycisphaerae bacterium

? GspH N_methyl T2SSG N_methyl T2SSF T2SSF T2SSE MshEN Arm-DNA-bind_3 Phage_int_M Phage_integrase PhdYeFM_antitox ParE_toxin HigB_toxin ?

JAFGTD010000010.1:1-13346
Acidobacteria bacterium

Amino_oxidase DUF169 OPT ParE_toxin DUF6290 Lactamase_B Glyco_hydro_31_2nd Glyco_hydro_31_3rd ? UxaE DctM DctQ

JAFGRW010000100.1:488-11522
Spirochaetes bacterium

? Bax1-I ? RelB-like HTH_19 ParE_toxin RelB-like MCPsignal HAMP sCache_2 ? Big Big MFS_1 Bact_hydrolase AAT TonB_C

JAFGWX010000060.1:10052-17062
Methylothermaceae bacterium

PALP ? DUF1318 TRAM tRNA_U5-meth_tr YkuD UPF0114 RelB-like ParE_toxin GIIM

JAFITR010000011.1:11733-18088
Simkania negevensis

GDH_HM3 Bac_GDH_CD Bac_GDH_CD ELFV_dehydrog Hydrolase E1-E2_ATPase VIT1 ? ParE_toxin GDC-P GDC-P GcvP2_C Aconitase Aconitase_C LicD ? Glyco_transf_9 TLC

JAFOLH010000003.1:185825-200473
Renibacterium sp.

Aldedh Sugar_tr Cupin_2 HTH_18 LssY_C ? SAICAR_synt GARS_N GARS_A GARS_C ParE_toxin PhdYeFM_antitox Asparaginase_II SCP_3 TctB Pribosyltran GATase_6 AIRS_C AIRS AbiEi_4

JAANWY010000387.1:710-7069
Anaerolineales bacterium

APH ParE_toxin PhdYeFM_antitox UPF0029 DUF1949 TPP_enzyme_C Fer4_7 EKR POR PFOR_II POR_N

JAFDXN010000012.1:135234-143161
Bacteroidetes bacterium

? Por_Secre_tail DUF6089 NAD_kinase NAD_kinase_C Beta-prop_EML_2 Cytochrom_D1 PSCyt1 FolB FGAR-AT_linker AIRS AIRS_C AIRS AIRS_C ParE_toxin RelB-like ?

JAGVYZ010000020.1:11020-22739
Candidatus Woesearchaeota archaeon

Fic KilA-N ? ? Yip1 ? DUF7557 ParE_toxin tRNA_SAD tRNA-synt_2b HGTP_anticodon ? ? FmrO ? Flavin_Reduct S-AdoMet_synt_N S-AdoMet_synt_M S-AdoMet_synt_C

JABJOR010000579.1:1-3919
Rhodospirillales bacterium

MTTB MTTB HEPN_AbiU2 PhdYeFM_antitox ParE_toxin

JAHHPH010000011.1:144102-12534
Verrucomicrobiia bacterium

? ? AMP-binding AMP-binding_C PP-binding Thioesterase ? ? ? ParE_toxin RelB-like ? Phage_integrase Phage_int_SAM_4 MYG1_exonuc Glyco_transf_5 Glyco_trans_1_4 COX2 DUF4083

JAHHMP010000021.1:7770-22027
Deltaproteobacteria bacterium

DUF1330 Bac_luciferase adh_short ADH_N ADH_zinc_N_2 DUF4437 ? DUF6290 ParE_toxin AA_permease_2 TrkA_N TrkA_C p450 TetR_N Rieske KshA_C MGH1-like_GH MGH1-like_GH

JAHJDR010000326.1:1736-3509
bacterium

? AbiEi_2 ParE_toxin DUF6290 Pribosyltran Peptidase_M23

JAHJOU010000072.1:3097-7481
Bacteroidetes bacterium

CoA_transf_3 ParE_toxin RelB-like HMGL-like Iso_dh POR TPP_enzyme_C POR_N PFOR_II Fer4_7 Fer4_16

JAFASY010000188.1:24867-30479

Chroococcidiopsidaceae cyanobacterium CP_BM_ER_R8_30

? ? XisI XisH Uma2 ParE_toxin ? ParE_toxin ? MazE_antitoxin ? ? ?

JAFDLV010000375.1:405-1992
Deltaproteobacteria bacterium

ParE_toxin PhdYeFM_antitox DUF2283 YafQ_toxin DUF6364

JAHHIK010000047.1:44535-59007
Tolypothrix carrinoi HA7290-LM1

Val_tRNA-synt_C CAAD Anticodon_1 tRNA-synt_1 PDDEXK_3 MFS_1 BcrAD_BadFG ? ? ? ParE_toxin ? PIN PhdYeFM_antitox ? ? ? tRNA-synt_2d

JAHHHL010000059.1:8557-23334
Lyngbya sp. HA4199-MV5

Peptidase_S9 PD40 ? FGase HemY_N TPRs TPRs TPR_MalT TPRs TPRs TPRs RNaseH-CATRA+Caspase ? ParE_toxin UPF0150 ParE_toxin LRRs LRRs LRRs LRRs LRRs REase RuBisCO_large_N RuBisCO_large RbcX RuBisCO_small Lipase_GDSL_2 AAA RuBisCO_activase_AAA_helical RuBisCO_small

JAHBUR010000012.1:73309-89451
Bacteroidetes bacterium

Abhydrolase_1 Calcineurin Na_Pi_cotrans Na_Pi_cotrans MS_channel_1st MS_channel_2nd MS_channel_3rd CitMHS CitMHS LapA_dom Radical_SAM CofH_C ParE_toxin PhdYeFM_antitox Radical_SAM CofH_C VitK2_biosynth ? ? Pan3_PK AAA_23 ORF6N

JAHBTI010000010.1:54923-69893
Bacteroidetes bacterium

VitK2_biosynth PNPase DUF1015 2-Hacid_dh_C GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C HTH_26 HKD ParE_toxin ? ? QueG_DUF1730 Fer4_16 PHA_synth_III_E Abhydrolase_1 POTRA POTRA ShlB Por_Secre_tail

JAHBTA010000002.1:674243-688625
Microbacteriaceae bacterium

Ribonuc_red_sm ATP-cone Ribonuc_red_lgN Ribonuc_red_lgC MFS_1 MFS_1 ABC2_membrane_3 ABC2_membrane_3 TetR_C_6 TetR_N Abhydrolase_6 Aldo_ket_red ParE_toxin PhdYeFM_antitox Abhydrolase_1 DUF4349 tRNA_SAD HpcH_HpaI HpcH_HpaI C-C_Bond_Lyase MaoC_dehydratas Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1

JAHBWO010000055.1:6770-19601
Nitrospira sp.

Sua5_C Sua5_yciO_yrdC Transposase_20 DEDD_Tnp_IS110 DUF6982 ODP Bestrophin-like TPRs TPRs ScsC_N DSBA ParE_toxin PhdYeFM_antitox TPM_phosphatase ? FAD_binding_3 HKD ? ThiS

JAIOHW010000998.1:1-2128
bacterium

ParE_toxin PhdYeFM_antitox MIP Apiosidase Collagen_bind_2

JAIQGH010000149.1:5528-7893
Terriglobia bacterium

? ? UbiD_N UbiD UbiD_C PhdYeFM_antitox ParE_toxin ? DDE_Tnp_IS66_C ?

JAIQGA010000082.1:1-11849
Terriglobia bacterium

DDE_Tnp_1 ? ParE_toxin PhdYeFM_antitox Rhodanese Rhodanese TPRs TPRs TPRs TPRs ? TMP-TENI Lyase_1 ADSL_C Lactamase_B_2

JAIQFW010000066.1:16406-22264
Terriglobia bacterium

? ? ? 2-oxoacid_dh E3_binding Biotin_lipoyl Cytochrome_CBB3 Transketolase_C Transket_pyr PhdYeFM_antitox ParE_toxin E1_dh

JAIVJQ010000010.1:1-4518
Acidobacteria bacterium

Sigma70_ECF STY-kinase TPRs TPRs ? RelB-like ParE_toxin ?

JAGQGS010000001.1:11426-18161
Thermomicrobiales bacterium

adh_short_C2 ? Uma2 Rieske Amidohydro_2 YjgF_endoribonc Aminotran_5 ParE_toxin PhdYeFM_antitox PhdYeFM_antitox

JAGQXO010000659.1:6611-20727
Phaeodactylibacter sp.

? ? OMP_b-brl_3 Plug CarboxypepD_reg His-Kinase His-Kinase Trans_reg_C Receiver DUF3958 ParE_toxin ? UbiA Por_Secre_tail MAM PKD_4 MAM Peptidase_M43

JACKEV010000008.1:705356-720207
Alphaproteobacteria bacterium

Peptidase_S9 Peptidase_S9_N Zn_ribbon_TFIIB Zn_ribbon_TFIIB Usp Creatininase Usp ? Cu-binding_MopE Epimerase Cupin_2 ? ParE_toxin ? Lambda_Bor FG-GAP FG-GAP FG-GAP FG-GAP FG-GAP_3 FG-GAP_3 MASE1 His-Kinase His-Kinase Acetylase ? ?

JAJHQX010000020.1:21514-28610
Armatimonadota bacterium

Methyltransf_4 Uma2 Amidase TPK_catalytic SteA-like_C MctB AAA_3 AAA_lid_2 ? ParE_toxin

JADLAP010000119.1:6760-14132
Phycisphaeraceae bacterium

? PhdYeFM_antitox ParE_toxin LacI Peripla_BP_3 N_methyl ? DUF4838 CBM9_1 ? GH123_N GH123_N LamininG LamininG

JAIORC010000072.1:1-2959
Phycisphaerae bacterium

PHP PALP HicB_lk_antitox HicA_toxin PhdYeFM_antitox ParE_toxin

JAGHCU010000094.1:3195-4914
Candidatus Latescibacterota bacterium

FlgD_ig Glyco_transf_9 ? ParE_toxin NTP_transf_2 ? ?

JAGWBM010000037.1:86248-98571
Pseudomonadales bacterium

CBM9_1 Calcineurin HTH_24 AsnC_trans_reg AlaDh_PNT_C AlaDh_PNT_N ? Pyr_redox_2 Fer4_20 NAD_binding_8 GATase_2 Glu_syn_central Glu_synthase GXGXG ParE_toxin DUF6290 TauD

QWGW01000019.1:23410-39738
Haliscomenobacteraceae bacterium CHB4

? TPRs Sigma70_r4 Sigma70_r2 OmpA DUF937 ? MRM3-like_sub_bind SpoU_methylase MazG MazG ParE_toxin Ribosomal_S15 ? RNase_PH RNase_PH_C PNPase RNase_PH RNase_PH_C KH_1 S1 PKD_4 Por_Secre_tail ? RoxA-like_Cyt-c CCP_MauG SnoaL

JAKRSB010000040.1:16047-15963
Thermosynechococcaceae cyanobacterium MS004

HTH_24 SufBD_N SUFBD_core Uma2 Transglutaminase Ig-like_KY ABC_tran SufBD_N SUFBD_core PhdYeFM_antitox ParE_toxin Aminotran_5 PAD_porph CN_hydrolase LPG_synthase_TM Pps Pps SWIM TP_methylase CbiG_C CbiG_N

JAFLSQ010000019.1:1249-8449
Treponema sp.

? ParE_toxin DUF6290 Ribosomal_S21 GntR FCD Cupin_2 HTH_18 ABC_tran ABC_membrane ABC_tran ABC_membrane GH43_C2 Glyco_hydro_43

JADFQN010000138.1:1-8323
Bacteroidetes bacterium

ShlB Lactamase_B_2 Methyltransf_25 TauE OsmC Peptidase_M20 MTHFR ParE_toxin RelB-like MTHFR ? COX4_pro

JACZRV010000007.1:17317-31635
SAR324 cluster bacterium

? FCSD_central Pyr_redox_2 ? HicB_lk_antitox B12-binding MeaB MM_CoA_mutase MM_CoA_mutase DJ-1_PfpI MerR_1 MerR-DNA-bind ParE_toxin PhdYeFM_antitox ATP-synt_ab_N ATP-synt_ab ATP-synt_VA_C ATP-synt ATP-synt_ab_N ATP-synt_ab ATP-synt_ab_C OSCP ATP-synt_B ATP-synt_B

JAKEFM010000023.1:1-11067
Planctomycetota bacterium

Fic PhdYeFM_antitox PIN Peptidase_M64 M64_N ParE_toxin PhdYeFM_antitox SLBB_2 SLBB Poly_export Glycos_transf_2 Glycos_transf_2 Glycos_transf_1 Glyco_transf_4 ? PMT_2

JAKEFD010000147.1:1-10457
Gemmataceae bacterium

STY-kinase HD_5 GAF_2 FHA DUF4139 ParE_toxin ? AP_endonuc_2 ? ? Peptidase_M48 Peptidase_M48_N Polysacc_deac_1

JALHUF010000027.1:1-4525
Terriglobia bacterium

BipA_C EFG_C EFG_III GTP_EFTU_D2 ? HTH_3 HigB-like_toxin ParE_toxin ? ?

JALHWK010000648.1:1-9110
Sedimentisphaerales bacterium

ParB_N HTH_Tnp_1_2 LacAB_rpiB PhdYeFM_antitox ParE_toxin Terminase_6C HTH_3 Peptidase_M78 ? Hydrolase_4 GCE_fung ?

JALHYD010000467.1:742-4291
bacterium

HicA_toxin HicB_lk_antitox ? DUF7387 HicA_toxin ? RelB-like ParE_toxin Caspase

JAGLZA010000201.1:4329-17517
candidate division WOR-3 bacterium

TEN_YD-shell ? ? F-box PIN ? RelB-like ParE_toxin RHS DUF192 DeoC SMC_N CHASE2 Guanylate_cyc B12-binding Radical_SAM NurA

JAGLYY010000127.1:12572-18902
Spirochaetales bacterium

DUF34_NIF3 Peptidase_A8 DUF2232 ? ? ? RelB-like ParE_toxin Fn3-I_PB4

JAGLTD010000388.1:1-4874
Spirochaetes bacterium

Cas_Cas7 Cas_TM1802 Polbeta ParE_toxin RelB-like Cas6b_N Cas6b_C CBM_6

JAIQZM010000030.1:30945-45699
Argonema galeatum A003/A1

Peptidase_M23 Fer4_5 Fer4_5 AAA_lid_14 NtrC-AAA cNMPBD DUF488 DUF488 DuOB ? ? ParE_toxin PhdYeFM_antitox N6_N4_Mtase ? ? Uma2 PhdYeFM_antitox Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans

JAMCWE010000265.1:13123-23748
Bacteroidetes bacterium

GWxTD_dom 2-Hacid_dh_C PGDH_inter ACT Thioredoxin adh_short_C2 DUF6883 DUF4926 PHP DNA_pol3_alpha DNA_pol3_finger HHH_6 N-OB ParE_toxin RelB-like Polbeta ?

JAGTJV010000004.1:21802-32677
Gloeotrichia echinulata DEX184

DUF2281 PIN Uma2 CPSF100_C Uma2 DUF29 ? DUF2281 ParE_toxin DUF2281 PIN HTH_Tnp_4 DDE_Tnp_4 Glycos_transf_2 Acyl_transf_3 ? ?

JAAEOU010002386.1:248-6903
bacterium

AlkZ-like NUDIX N6_Mtase N6_Mtase XisH XisI ? RelB-like ParE_toxin REase ? DeoRC HTH_DeoR ROK

JAAEQP010000826.1:3343-16903
bacterium

GFO_IDH_MocA DUF362 Methyltransf_25 PQQ_2 PQQ_2 PQQ_2 PQQ_2 PQQ_2 PhoD_N DUF2341 LamininG Hydrolase_6 Hydrolase_like HAD_2 PhdYeFM_antitox ParE_toxin UvrD_C UvrD_C UvrD-helicase BD-FAE

JALOLV010000297.1:358-4220
Spirosomataceae bacterium

SSF ClpS RecR_HhH RecR_ZnF Toprim_4 RecR_C PRAI ? ParE_toxin

JALOOZ010000146.1:1-11550
Thermodesulfobacteriota bacterium

? HlyD_D23 ABC_tran FtsX FtsX MacB_PCD DUF169 ParE_toxin PhdYeFM_antitox MqsA_antitoxin MqsR_toxin CoA_binding_2 Succ_CoA_lig Ligase_CoA_2 ETF ETF_alpha

JAODBD010000003.1:44766-57717
Glaciihabitans sp.

PP2C_2 Yop-YscD_cpl FhaA_N FHA COX15-CtaA NAD_binding_10 MarR_2 NMN_transporter ParE_toxin PhdYeFM_antitox ? ? ? ? ? VanZ

JAPLMN010000105.1:3632-18930
Planctomycetota bacterium

Trypsin PDZ_2 PDZ ArsP_1 ? TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs Alginate_lyase ParE_toxin PhdYeFM_antitox ADH_zinc_N ADH_N PHA_gran_rgn ? ? ACT Homoserine_dh NAD_binding_3 PhosphMutase

JAPLOD010000808.1:1-7542
Planctomycetota bacterium

Beta-prop_ACSF4 PQQ_2 PQQ_2 TPRs zf-HC2 Sigma70_r2 Sigma70_r4_2 ? ParE_toxin PhdYeFM_antitox SRAP

JAPUJJ010000338.1:3904-4232
Acidobacteria bacterium

PhdYeFM_antitox ParE_toxin ? ? ?

JAQCNZ010000020.1:151015-159956
Chloroflexota bacterium

ABC_tran TOBE_2 BPD_transp_1 BPD_transp_1 SBP_bac_8 23S_rRNA_IVP Aminotran_3 ? ParE_toxin ? Lysyl_oxidase

JAPTXZ010000265.1:400-2486
Actinomycetota bacterium

? PhdYeFM_antitox ParE_toxin F420H2_quin_red

JAJRDX010000067.1:786-8715
Methylococcaceae bacterium

ParE_toxin DUF6290 LptF_LptG LptF_LptG Sel1 Sel1 Sel1 Sel1 His_Phos_1 CHAD Gly-zipper_YMGG OmpA

JAQUAZ010000127.1:40-4937
Methylococcales bacterium

ParE_toxin ? ? ? Zot DUF5455 ? Phage_coatGP8

JAQUQV010000013.1:19571-38944
Kiritimatiellae bacterium

tRNA-synt_1b CBM_48 Alpha-amylase Alpha-amylase_C AMPK1_CBM Amidohydro_1 ? NAD_binding_2 6PGD DUF4838 CBM_4_9 ParE_toxin PhdYeFM_antitox TPRs POR TPP_enzyme_C POR_N PFOR_II Fer4_7 ADH_N ADH_zinc_N

JAPPEK010000605.1:3277-14246
Gemmatimonadota bacterium

HTH_20 CarbopepD_reg_2 Plug TonB_dep_Rec_b-barrel DUF4249 DUF5128 DUF3667 ? DUF6290 ParE_toxin ? ? Sortilin-Vps10 Sortilin-Vps10

JAOTYS010000732.1:87-2064
Gammaproteobacteria bacterium

RHH_1 DUF6290 ParE_toxin Ysc84 ?

JASJDW010000001.1:17146-31227
Crocosphaera sp.

GST_C_2 GST_N_2 DUF6887 DUF6888 FitA-like_RHH ? PIN HicB_lk_antitox ParE_toxin PhdYeFM_antitox GDE_N GDE_C Peptidase_M16_C Peptidase_M16 Peptidase_M16 Receiver His-Kinase His-Kinase GAF PAS PAS GAF CBS CBS CBS CBS Coprogen_oxidas ?

JASJDI010000331.1:1-6773

Xenococcaceae cyanobacterium MO_188.B29

Receiver ? GNVR Poly_export SLBB_2 Glyco_transf_61 ParE_toxin RelB-like HEPN_MAE_28990

JASJCT010000022.1:172960-192690
Myxococcota bacterium

Patatin PC4 AIRS_C AIRS AIRS_C AIRS FGAR-AT_linker GATase_5 HTH_17 ABC-ATPase CCDC113_CCDC96_CC AAA_15 HATPase_c_4 SLFN_AlbA_2 ParE_toxin PhdYeFM_antitox ? Ribonucleas_3_3 dsrm Thioredoxin_4 VKOR GreA_GreB GreA_GreB_N RecG_dom3_C Helicase_C DEAD RecG_wedge HEATs SGSH_C Sulfatase

JASJCJ010000090.1:9061-29683

Calothrix sp. MO_167.B12

Epimerase PUA_4 Methyltrans_RNA TPRs PsiE TPRs TPRs TPRs TPRs GTP_EFTU EF-G_D2 EFG_III EFG_IV EFG_C ? ParE_toxin ? GUN4 STY-kinase STY-kinase zf-FPG_IleRS Anticodon_1 tRNA-synt_1 ? RNaseH-CATRA+Caspase AP-ATPase FGS Uma2

JASJBX010000119.1:12882-5501
Crocosphaera sp.

HTH_37 HicB_lk_antitox ? ? ? ParE_toxin ? ? PaaA_PaaC Uma2 Host_attach Cu-oxidase_4 DUF6887

JAUSFG010000340.1:294-4148
Phycisphaerales bacterium

PhdYeFM_antitox ParE_toxin VTT_dom Vut_1 ? VWA

JALYUK010000495.1:1-2739
Actinomycetota bacterium

APH ? PhdYeFM_antitox Unstab_antitox ParE_toxin ?

JALZIH010000398.1:1-4809
Acidobacteriota bacterium

? ? ? ParE_toxin ? Peptidase_M15 EPSP_synthase SKI

JALZIH010000441.1:1-3749
Acidobacteriota bacterium

YidC_periplas YidD Ribonuclease_P Ribosomal_L34 ParE_toxin ? Formyl_trans_N CarboxypepD_reg ?

JAISJO010000023.1:1-8626
Propionibacteriaceae bacterium

BRCT ParE_toxin PhdYeFM_antitox Phage_integrase SBP_bac_8 Alpha-amylase ABC_tran DrrA1-3_C ? Zn_ribbon_9 CT398_CC

JAISAC010000193.1:2128-8326
Prevotellaceae bacterium

OEP OEP DUF4861 GATase_2 GATase_6 DUF4160 DUF2442 DUF3802 ? ParE_toxin

JAITCS010000155.1:1135-6045
Cellulomonadaceae bacterium

FAD_syn ? ParE_toxin PhdYeFM_antitox GerE ? ?

JAITOZ010000260.1:1-7922
Planctomycetaceae bacterium

T2SSE FMN_red Beta-prop_THOC3 SHD1 DUF7382 ? ParE_toxin ? DUF4139

JAXHOK010000050.1:7481-17785
Arcicella sp.

dCMP_cyt_deam_1 DUF559 Ceramidase_alk Oxidored_q4 Oxidored_q6 Complex1_30kDa ? ParE_toxin NiFeSe_Hases Complex1_49kDa ? DUF29 2Fe-2S_thioredx PhyH ? Complex1_51K SLBB_2 NADH_4Fe-4S

JAVRCF010000002.1:170289-185172
Aulosira sp. DedQUE10

Acetyltransf_7 GST_C GST_N_3 AhpC-TSA 1-cysPrx_C AhpC-TSA GDE_N GDE_C Peptidase_M24 ? ParE_toxin Cupredoxin_1 Peptidase_M16_C Peptidase_M16 Peptidase_M16_C Peptidase_M16 Glyoxalase DUF2396 Hexapep Hexapep PsbY Mur_ligase_M Mur_ligase_C

JAVRCG010000013.1:155440-174027
Nostoc sp. DedQUE11

Ribosomal_S6 DUF641 TMF_TATA_bd STAS VTT_dom Arginosuc_synth Arginosuc_syn_C TPRs TPRs TPRs TPRs NACHT RNaseH-CATRA+Caspase RNaseH-CATRA+Caspase ? ParE_toxin ? ? ? PhoX PhoX Glyco_trans_2_3 Glycos_transf_2 Peptidase_S66 Peptidase_S66C

JAYEHU010000114.1:43-4467
Planctomycetota bacterium

ParE_toxin ? SBP_bac_10 N_methyl BFN_dom ACT_9 ACT ACT_9 ACT_9 AA_kinase

JAYEMF010000019.1:15315-23086
Campylobacterota bacterium

FIST FIST_C PAS His-Kinase His-Kinase PAS HAMP_2 MCPsignal PAS GAF PAS His-Kinase His-Kinase Receiver GerE RHH_5 ParE_toxin HepT-like

JAYKJI010000224.1:725-7592
Snowella sp.

WYL WCX Cas6 Cas3_HD DEAD Cas3-like_C_2 ? DevR RHH_4 ParE_toxin Cas_Cas5d

JAYKJI010000431.1:5001-15257
Snowella sp.

STY-kinase FGS 23S_rRNA_IVP ? ? BrnT_toxin BrnA_antitoxin AIPR ParE_toxin DUF2281 XisI STY-kinase FGS Fer4_12 ?

JAYKJC010000189.1:1-2780

Leptolyngbyaceae cyanobacterium

TPRs TPRs RelB-like ParE_toxin HicB_lk_antitox Tc-R-P Methyltransf_3

JAYKIC010000036.1:1454-6157

Leptolyngbyaceae cyanobacterium

Epimerase ? ? Excalibur SNase Glucokinase ? ParE_toxin DUF2281 DUF2281 ?

JAYKGY010000050.1:932-7709
Snowella sp.

Cas3_HD DEAD Cas3-like_C_2 ? BrnT_toxin Cas8b_N Cas8b_C DevR Cas_Cas5d RHH_1 ParE_toxin

JAWQFP010000179.1:4547-4383
Scytonema sp. PMC 1069.18

PIN Uma2 ? ParE_toxin ? Calcineurin ? ?

JAWQFU010000058.1:165-7781
Oscillatoria sp. PMC 1050.18

HIT ChlI Mg_chelatase Mg_chelatase_C ? ParE_toxin Lyase_1 ADSL_C Ni_hydr_CYTB ? DUF423 ? Lactamase_B_3 Uma2 HTH_7

JAWQFX010000217.1:1-8025
Oscillatoria sp. PMC 1076.18

Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans PhdYeFM_antitox ParE_toxin Pps Pps Pps Pps Pps Pps Pps Pps Pps Pps Pps Pilin_GH LTXXQ ?

QGUM02000135.1:1-4491
Actinomycetes bacterium

Sod_Fe_C Sod_Fe_N ParE_toxin PhdYeFM_antitox adh_short HSDR_N_2 Cyt_c_ox_IV COX1

SIAS01000005.1:44330-50050
Verrucomicrobiia bacterium

? Thioredoxin Cys_rich_CPXG DUF2064 GHMP_kinases_C GHMP_kinases_N Ldh_1_N Ldh_1_C ? ParE_toxin ANKs ANK_LRRK2 ANKs

JAAAOF010000278.1:48-7279
Bacteroidetes bacterium

Receiver HTH_18 Receiver His-Kinase His-Kinase Y_Y_Y Reg_prop Reg_prop Reg_prop Reg_prop Reg_prop Reg_prop ParE_toxin PhdYeFM_antitox Phage_integrase

JAAHGY010000050.1:15356-14760
Cyanothece sp. SIO2G6

CRISPR_Cas2 Cas_Cas4 Cas_Cas1 DUF7340 HNH ? HrcA MYO10_CC Uma2 Cas_Cas5d ParE_toxin ? DevR ? ? Cas3_HD DEAD Cas3-like_C_2 Cas6 HNOBA Guanylate_cyc

JAAHGY010000434.1:42-5924
Cyanothece sp. SIO2G6

? PhdYeFM_antitox ParE_toxin PPC TPRs TPR_NPHP3 TPR_MalT TPRs TPRs TPRs TPRs TPRs RNaseH-CATRA+Caspase

JAAKFX010000055.1:1-10326
Candidatus Anoxychlamydiales bacterium

HTH_16 RNA_pol_Rpb2_6 RNA_pol_Rpb2_7 RNA_pol_Rpb1_1 RNA_pol_Rpb1_2 RNA_pol_Rpb1_2 RNA_pol_Rpb1_3 RNA_pol_Rpb1_4 RNA_pol_Rpb1_5 MFS_1 ? TAL_FSA PhdYeFM_antitox ParE_toxin ?

JAACEL010000445.1:197-3689
Nitrospiraceae bacterium

peroxidase peroxidase ParE_toxin RelB-like

JAAUUQ010000134.1:5058-6680
bacterium

NACHT HEATs RelB_like_antitoxin ? ? ParE-like_toxin DUF2281 ParE_toxin CTP_transf_1 RRF DDE_Tnp_IS1

JAAUPN010000074.1:5944-17165

Coleofasciculaceae cyanobacterium RL_1_1

? SpoIIE Arginase MethyltransfD12 MethyltransfD12 DUF3747 Proton_antipo_N Proton_antipo_M Proton_antipo_C ParE_toxin DUF2281 Hexapep

JAAUPR010000046.1:17041-28729

Limnothrix sp. RL_2_0

? Nucleos_tra2_N Gate Nucleos_tra2_C MNS-STAND Beta-prop_THOC3 Beta-prop_THOC3 Beta-prop_WDR3_1st MNS-STAND GGDEF MNS-STAND DUF4335 DUF3379 DUF3038 ParE_toxin DUF2281 GIY-YIG ?

JABFSO010000050.1:1-7949
Cyclobacteriaceae bacterium

Peptidase_M3 Peptidase_M3 ParE_toxin RelB-like MarR YceI CBM9_1 DUF5916 MTS Fe-ADH ADH_Fe_C AbiEii AbiEii

JABFSN010000010.1:5066-8050
Phycisphaerales bacterium

AcetylCoA_hyd_C Acetylase DUF4062 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 PhdYeFM_antitox ParE_toxin DUF7560 ?

JAADIN010000091.1:1863-5155
Planctomycetota bacterium

PhdYeFM_antitox ParE_toxin Phage_integrase ? UPF0102 Ribosomal_L19 tRNA_m1G_MT Ribosomal_S16 ParB_C

JABMQP010000080.1:113-8403
Candidatus Aminicenantes bacterium

? Methyltransf_31 HTH_3 RHH_3 ParE_toxin HicB_lk_antitox HicA_toxin NTP_transf_2 PRiA4_ORF3 RNaseH-CATRA+Caspase Beta-lactamase DUF554

JABMMJ010000277.1:1-2860
bacterium

Ribosomal_L27 OpcA_G6PD_assem OpcA_G6PD_C ParE_toxin RelB-like G6PD_C

JAAXXD010000031.1:203-10574
Chlorobaculum sp.

GAF NtrC-AAA AAA_lid_14 HTH_8 DUF1643 RelB-like ParE_toxin Oxidored_molyb TonB_dep_Rec_b-barrel Plug ABC_tran cobW SBP_bac_6 Peripla_BP_2 Methyltransf_11

JAAXXD010000118.1:307-4294
Chlorobaculum sp.

DUF4062 MqsA_antitoxin ParE_toxin ?

JAATWA010000001.1:97086-109656
Pseudanabaena biceps

AAA_23 Rad50_zn_hook Tup_N SbcC_Walker_B PemK_toxin ? TPRs TPRs TPRs STY-kinase ? ? ParE_toxin Radical_SAM LIAS_N ? DUF3318 TPRs TPRs TPRs TPRs Zn_ribbon_LapB ATPase ChlI ? NinF PALP

MEKR01000004.1:821-11067

Acidobacteria bacterium RIFCSPLOWO2_02_FULL_61_28

PQQ PQQ ABC_tran Receiver NtrC-AAA AAA_lid_14 HTH_8 ParE_toxin RelB-like HisKA_7TM GAF_2 PAS His-Kinase His-Kinase ABC_tran Lactamase_B HicB_lk_antitox

MEKY01000062.1:15241-12654

Acidobacteria bacterium RIFCSPLOWO2_12_FULL_59_11

? ? 5_3_exonuc_N 5_3_exonuc DNA_polI_exo1 DNA_pol_A PIN PhdYeFM_antitox ParE_toxin LolA_like ? ABC_tran FtsX MacB_PCD LolA_like ? CarboxypepD_reg Plug OMP_b-brl_4

MGBU01000112.1:9275-9273

Candidatus Rokubacteria bacterium GWA2_73_35

ParE_toxin ? DUF6941 EamA EamA AFOR_N AFOR_C TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs Hydrolase

MGPV01000013.1:25807-30850

Deltaproteobacteria bacterium GWC2_65_14

HicB_lk_antitox PIN PIN MazE_antitoxin ? PIN RHH_1 ParE_toxin PhdYeFM_antitox AAA_14 DUF4143

MGRF01000023.1:459-6821

Deltaproteobacteria bacterium RBG_19FT_COMBO_60_16

Na_H_Exchanger DUF3303 ParE_toxin PhdYeFM_antitox ORF6N DoxX HTH_3 ? His_biosynth

MGRG01000008.1:186274-200671

Deltaproteobacteria bacterium RIFCSPHIGHO2_01_FULL_43_49

T2SSE MshEN POR POR_N TPP_enzyme_C PFO_beta_C CSD Omp85 CN_hydrolase DUF6290 ParE_toxin ? Hydrolase_4 DUF433 DUF5615 KARI_C KARI_N ? Anticodon_2 tRNA-synt_1c

MGRR01000041.1:8200-7364

Deltaproteobacteria bacterium RIFCSPHIGHO2_12_FULL_43_9

? PhdYeFM_antitox ParE_toxin Transketolase_N Transket_pyr Transketolase_C dnstrm_HI1420 Gp49 Lipase_GDSL ThiI THUMP ThiI_fer Aminotran_5

MGST01000313.1:16361-28618

Deltaproteobacteria bacterium RIFOXYA12_FULL_58_15

AbiEii SNase LTD SNase Fer4_20 Pyr_redox_2 DHO_dh Fer4_21 His-Kinase PilZ ? ParE_toxin XPB_DRD ResIII Helicase_C DUF790 PPDK_N PhdYeFM_antitox

MHXR01000049.1:53153-67983

Planctomycetes bacterium GWA2_50_13

CN_hydrolase NAD_synthase ? HD ? Cas_Cas1 CRISPR_Cas2 ? ParE_toxin Acyltransferase S1 S1 S1 S1 S1 S1 Sigma70_r4 Sigma70_r2 Sigma70_r1_2 Pribosyltran Peptidase_S24 HTH_3 Amidohydro_3 Cytidylate_kin

LWDK01000048.1:38915-51134

Proteobacteria bacterium SG_bin7

ICMT ? ? ? ? RWP-RK Recombinase PAD ParE_toxin RH3_dom PSCyt1 YceI ? ? ? CRCB

NBMI01000049.1:27722-44302

Candidatus Parabeggiatoa sp. nov. 2

RuBisCO_small CbiA LysM SH3_3 ? ? ? ParE_toxin ? ? AAA_15 DUF4435 HSP70 Nuf2_DHR10-like Fer4_12 FGS

PEZS01000001.1:218413-229160

Bdellovibrionales bacterium CG10_big_fil_rev_8_21_14_0_10_45_34

NTase_sub_bind NTP_transf_2 ? ParE_toxin PhdYeFM_antitox HTH_3 ParE_toxin DUF6290 ? ? Gp49 HNH Cofac_haem_bdg Cyt_bd_oxida_II Gp49

PVWL01000028.1:45416-55668
Cyanosarcina cf. burmensis CCALA 770

Xre_MbcA_ParS_C Xre-like-HTH RES ? HTH_23 ? ? RHH_1 ParE_toxin PhdYeFM_antitox HTH_26 ? Polbeta HEPN ? ?

PXSZ01000044.1:73375-82073

Bacteroidetes bacterium QH_10_64_37

Proton_antipo_M Oxidored_q5_N Pterin_4a Proton_antipo_M ? Aldedh Isochorismatase HicB ParE_toxin RelB-like DUF5615 DUF433 DUF433

PXTI01000061.1:1609-11573

Bacteroidetes bacterium QH_8_67_23

SecD_SecF_C Sec_GG STAS Metallophos_2 Amidohydro_1 ParE_toxin PhdYeFM_antitox Peptidase_M48 Rce1-like LysM LysM LysM DPBB_1 TPRs Trehalase Mannitol_dh Mannitol_dh_C

CP050884.1:3766572-3780402
Dolichospermum sp. DL01

Dala_Dala_lig_C Dala_Dala_lig_C LAM_C Radical_SAM AAA_26 DRTGG Ycf66_N NAD_binding_2 6PGD ParE_toxin DUF4276 HicA_toxin HicB_lk_antitox ? DUF2326 CTD10 AAA_23 MC6 ? PemK_toxin

QEVQ01000161.1:1-4807
Proteobacteria bacterium

Memo ? PhdYeFM_antitox ParE_toxin ?

QNEL01000080.1:216-20854
Candidatus Parabeggiatoa sp. nov. 3

HicB_lk_antitox Glycos_transf_2 Glycos_transf_2 Sulfotransfer_3 ? ? ParE_toxin Calx-beta PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 SPASM Radical_SAM Sulfotransfer_3 Sulfotransfer_1 Sulfotransfer_3 Glycos_transf_2 ? Sulfotransfer_3 AAA_23 ABC-ATPase HicA_toxin HNH

QMRP01000058.1:655-8737
Thermoprotei archaeon

FAD_binding_3 GGR_cat PIN ? ? DUF2283 DUF2683 ParE_toxin ParA HypA Calcineurin ?

RFIY01000040.1:13182-26060
Candidatus Dadabacteria bacterium

? SpoVG Receiver PilZ Na_Ala_symp BATS Radical_SAM IstB_IS21 RHH_1 ParE_toxin HAD Radical_SAM HrcA_DNA-bdg HrcA GrpE Peptidase_M20 Acetylase Pribosyltran Pribosyltran_N

RDXI01000010.1:38673-51877
Bacteroidetes bacterium

GldL_N FGS PorP_SprF HEM4 DUF4271 ? SUFBD_core SufBD_N ParE_toxin ? HTH_3 Methylase_S Methylase_S Big Big Por_Secre_tail

SHNH01000008.1:41157-52305
Candidatus Saccharibacteria bacterium

? Glyco_hydro_18 fn3 5_3_exonuc_N 5_3_exonuc DNA_pol_A Hydrolase MazE_antitoxin ? Antitox_RHH ParE_toxin ? Aminotran_1_2 Methyltransf_25 NUDIX PhdYeFM_antitox PIN ?

SMYC01000093.1:4226-19990
Acidobacteria bacterium

DUF4159 DUF6483 ? Trans_reg_C Sulfatase Sulfatase Beta-lactamase Metal_resist DUF6290 ParE_toxin Cu-oxidase_2 Cu-oxidase Cu-oxidase_3 ? ? Amidohydro_1 Abhydrolase_1 STY-kinase PEGA PEGA TonB_C Peripla_BP_2 ?

RDSB01000021.1:64779-72306
Marinifilum sp. JC120

Lar_restr_allev Phage_CP76 ? ? HTH_3 HTH_26 Peptidase_S24 ? ParE_toxin ? ? ? ? Lar_restr_allev PAPS_reduct ?

VBOP01000117.1:3425-3422
Nitrospirae bacterium

MazE_antitoxin PemK_toxin ParE_toxin PhdYeFM_antitox FecR

VIKD01000012.1:1-8939
Planctomycetota bacterium

Calx-beta Calx-beta ? BD-FAE ? ? HTH_3 ParE_toxin PhdYeFM_antitox ? MmeI_Mtase Peptidase_M78 ? ?

NZ_ANKQ01000001.1:1510034-1527054
Microcystis aeruginosa TAIHU98

Methylase_S Methylase_S Ribosomal_L27A NTP_transf_2 ? SIS SIS GATase_6 Ribosomal_L6 Ribosomal_L6 Ribosomal_L18p ParE_toxin N6_Mtase HsdM_N Ribosomal_S5_C Ribosomal_S5 ? REase SecY ADK Tnp_DNA_bind DDE_Tnp_1 Dimer_Tnp_Tn5

NZ_AGQQ02000002.1:348234-361419
Corynebacterium glutamicum ATCC 14067

Hydrolase MFS_1 DeoRC HTH_DeoR FGGY_C FGGY_N Pantoate_ligase ? ? ParE_toxin PhdYeFM_antitox HTH_17 Pur_DNA_glyco DUF1801 HHH_5 Hexapep Hexapep MarR_2 ROK ROK Pantoate_transf

NZ_JQRQ01000010.1:3419-11963
Leptospira weilii

HTH_3 ? ? ParB_N ParA-Soj-NTPase Phage_integrase Toprim_2 DNAG_N Zn_ribbon_DnaG RelB-like ParE_toxin

NZ_BAHD01000004.1:149258-170430
Kineosphaera limosa NBRC 100340

adh_short_C2 DUF1949 UPF0029 Cys_Met_Meta_PP Viral_helicase1 UvrD-helicase PNKP_ligase Calcineurin AAA_33 PhdYeFM_antitox ParE_toxin Rce1-like DUF480 Methyltransf_25 Catalase Catalase-rel FUR DUF3418 OB_NTP_bind HA2_C HA2_N Helicase_C DEAD DUF1206 DUF1206 DUF1206 Methyltransf_12 Hen1_L PAC2

NC_014502.1:103422-117840
Gloeothece verrucosa PCC 7822

Trypco1 ? ParE_toxin ? AAA ? TraG-D_C ? ? ? nSTAND1 Caspase Peripla_BP_6 ? DUF772 ?

NC_015572.1:3823796-3833744
Methylomonas methanica MC09

T6SS_Tsi2-like ADPRTs_Tse2 NERD Zn_ribbon_Top1 ? ? ? ParE_toxin DUF6290 ParE_toxin polyprenyl_synt DXP_synthase_N Transket_pyr Transketolase_C Pribosyltran MlaA Exonuc_VII_S ?

NC_018012.1:1679493-1693944
Thiocystis violascens DSM 198

HTH_3 ? YgjP-like REase SWI2_SNF2 T1R_D3-like ParE_toxin RelB-like Methylase_S HsdM_N N6_Mtase RHH_5 ? ? Arm-DNA-bind_3 Phage_integrase Phage_int_M DUF6941 ?

NC_019779.1:3408635-3423070
Halothece sp. PCC 7418

DNA_methylase ? S-methyl_trans Pterin_bind B12-binding_2 B12-binding Met_synt_B12 PAF-AH_p_II Abhydrolase_1 DUF1400 DHQS_C Fe-ADH Aminotran_1_2 Methyltransf_25 ParE_toxin ? HicB_lk_antitox HicA_toxin ? T5orf172 CopG_antitoxin ?

NC_012804.1:1040168-1051183
Thermococcus gammatolerans EJ3

ABC-3 RNase_HII GT87 Methyltransf_15 ? ParE_toxin eRF1_1 eRF1_2 eRF1_3 HEPN NTP_transf_2 AIRS_C AIRS QueC GPDPase_memb FeoA Creatinase_N Peptidase_M24 PUMA_CC

NZ_KE332518.1:1230510-1242007
Treponema maltophilum ATCC 51939

? DUF5692 Amino_oxidase FAD_binding_6 NAD_binding_1 Fer2 DUF5692 Acyltransferase DUF6290 ParE_toxin ? ? dsrm Ribonucleas_3_3 PP-binding Ribosomal_L32p BrnT_toxin BrnA_antitoxin

NZ_KB235933.1:412158-430686
Prochlorothrix hollandica PCC 9006 = CALU 1027

? BrxC_BrxD BrxC_BrxD DDE_Tnp_ISL3 HTH_Tnp_ISL3 Zn_ribbon_ISL3 ArAE_1_C ? Pyr_redox_2 AIRS AIRS_C NTP_transferase LbH_EIF2B SMC_ScpA ABC-ATPase ? Hexapep Hexapep Hexapep NTP_transf_3 DUF3226 DEAD Helicase_C WH_Lhr BrxC_BrxD ParE_toxin

NZ_KB235948.1:3848664-3864153
Kamptonema formosum PCC 10802

GATase_7 Asn_synthase DAGK_cat YegS_C PAS PAS PAS PAS T4_CASP His-Kinase PhdYeFM_antitox ParE_toxin PhdYeFM_antitox DUF6290 Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans ? YafQ_toxin Ead_Ea22 ACP_PD ? DUF433 MotA_ExbB TOP6A-Spo11_Toprim

NZ_KQ976354.1:3426810-3441193
Scytonema hofmannii PCC 7110

MoxR-AAA FGS bAvd RVT_1 ? ? ParE_toxin LysR_substrate HTH_1 ? ? Gly_kinase p450 AAA_16 Glyco_transf_28 ?

NZ_JAJNBZ010000027.1:5246-23883
Paenibacillus profundus

FecCD FecCD HlyD_2 Peripla_BP_2 ? ? ParE_toxin Asp_protease_2 ? HTH_18 Peripla_BP_2 N-OB tRNA-synt_2 Abhydrolase_1 Abhydrolase_1 SBP_bac_1 FlgM Beta-lactamase

NZ_ATWH01000005.1:33040-46544
Glaciibacter superstes DSM 21135

AP_endonuc_2 GFO_IDH_MocA GFO_IDH_MocA_C DUF6807 SBP_bac_8 BPD_transp_1 PhdYeFM_antitox ParE_toxin BPD_transp_1 SBP_bac_3 Histidinol_dh HTH_1 LysR_substrate OCD_Mu_crystall PEP_mutase ABC_tran BPD_transp_1

NZ_JIAG01000001.1:328022-342820
Haematomicrobium sanguinis DSM 21259

MFS_1 MFS_1 FemAB Acetyltransf_6 DoxX PhdYeFM_antitox DUF6290 ParE_toxin SAICAR_synt GARS_C GARS_A GARS_N ? UxaC Peripla_BP_3 LacI AP_endonuc_2 IclR_C HTH_IclR Sugar_tr GFO_IDH_MocA GFO_IDH_MocA_C3

NZ_ATYW01000002.1:367686-384492
Glycomyces tenuis DSM 44171

FMN_red ? BPD_transp_2 CarboxypepD_reg BPD_transp_2 ParE_toxin PhdYeFM_antitox adh_short ADH_N ADH_zinc_N MLTR_LBD HTH_31 Peripla_BP_6 ABC_tran Cas12f1-like_TNB OrfB_IS605 MFS_1 Trans_reg_C BTAD AP-ATPase TPR_MalT TPR_MalT ABC_tran BCA_ABC_TP_C

NZ_BAMD01000056.1:23913-35672
Saccharicrinis fermentans DSM 9555 = JCM 21142

ParE_toxin HTH_3 ParE_toxin RelB-like HTH_3 ? ? ? RelB-like Unstab_antitox ? Phage_integrase Phage_int_SAM_1 ? ? PH_LRR1 Phage_integrase

NZ_PCOW01000001.1:195675-215390
Microbacterium sp. MYb64

DUF1918 UPF0176_N Rhodanese Rhodanese_C VTT_dom DUF883_C ResIII AAA_lid_2 AAA_3 PhdYeFM_antitox ParE_toxin DUF222 HNH_2 ? Hydrolase Glyco_hydro_65N Glyco_hydro_65m Glyco_hydro_65C DNA_pol3_delta2 DNAX_ATPase_lid DNA_pol3_gamma3 RecR_HhH RecR_ZnF Toprim_4 RecR_C Transglutaminase TgpA_N vWA

NZ_LUUI01000127.1:101676-108783
Methylomonas lenta

DUF3368 HTH_3 Peptidase_M78 Methylase_S ? ParE_toxin Cation_efflux MerR-DNA-bind MerR_1 RelB-like Methylase_S ? REase SWI2_SNF2

NZ_JYIJ01000008.1:687-17911
Carbonactinospora thermoautotrophica

PhdYeFM_antitox ParE_toxin Phage_int_SAM_1 Phage_integrase LPP WXG100_2 ? ? Methyltransf_25 Eco57I ResIII Helicase_C ParA-Soj-NTPase

NZ_BCRE01000100.1:55-8300
Kribbia dieselivorans NBRC 106261

ADH_zinc_N_2 ADH_N ? Aldo_ket_red RibD_C ParE_toxin NTP_transf_2 MMR_HSR1 YchF-GTPase_C PhdYeFM_antitox PhdYeFM_antitox HTH_3 HepT-like PIN Aldo_ket_red

NZ_JACKZR010000813.1:23827-40657
Nostoc sp. UCD120

DUF3531 NatA_aux_su HemY_N NUDIX UPF0150 ? ParE_toxin ? UvrB_inter CarD_TRCF_RID DEAD Helicase_C TRCF CorC_HlyC CBS CBS CNNM Aldo_ket_red AMP-binding AMP-binding_C PP-binding Condensation AMP-binding AMP-binding_C PP-binding Thioesterase HicA_toxin ?

NZ_KV788212.1:32136-42564
Corynebacterium sp. HMSC05E07

ABC_tran Aminotran_5 ADH_N ADH_zinc_N Pyr_redox_2 ? ParE_toxin Sirtuin DUF1648 ? ABC2_membrane ThiF Rhodanese ? ? PhdYeFM_antitox

NZ_FQUS01000045.1:11796-7532
Fodinibius roseus

PhdYeFM_antitox ParE_toxin UDPG_MGDP_dh_C UDPG_MGDP_dh UDPG_MGDP_dh_N DUF2442 DUF4160 Epimerase_2 ? Glycos_transf_4 Glyco_tran_WecG

NZ_FTOP01000021.1:16432-30960
Belliella pelovolcani

? ? TolB_like ArsC Aminotran_3 Peripla_BP_6 RelB-like clamp_Gag1-like ParE_toxin RMMBL Lactamase_B ? Flavin_Reduct adh_short_C2 adh_short GMP_synt_C NAD_synthase GATase ? Strep_SA_rep

NZ_CAWRIF010000390.1:144-14925
Calothrix rhizosoleniae SC01

GATase CTP_synth_N Glucodextran_B SH3_3 Amidase_3 GmrSD_N HEPN_RiboL-PSP ? ParE_toxin MMR_HSR1 KH_2 ? EamA EamA ? ? DUF1802 Mrr_cat KAP_NTPase

NZ_FOLQ01000058.1:1-9856
Spirosoma endophyticum

Glycos_transf_1 Glyco_transf_4 STELLO ? DUF7683 ? ParE_toxin ? Rep3_N Rep3_C ? DUF5712 ? OMP_b-brl

NZ_NOLJ01000002.1:799522-810603
Nostoc sp. 'Peltigera membranacea cyanobiont' 210A

Receiver His-Kinase His-Kinase PAS PAS Receiver MMR_HSR1 YchF-GTPase_C ? YidD ? Pps Pps ParE_toxin SAP ? Pps Pps Pps Abhydrolase_6 AhpC-TSA AA_kinase ? ?

NZ_BMKX01000013.1:81976-95892
Glutamicibacter ardleyensis

PadR HAAS Resolvase HTH_7 YobI-ATPase UvrD-helicase UvrD_C SLFN_AlbA_2 HTH_7 Resolvase PhdYeFM_antitox ParE_toxin ? ABC-ATPase OLD-like_TOPRIM ?

NZ_LN867111.1:237-8877
Estrella lausannensis

? ? DUF5906 Zn_Ribbon_Prim Toprim_2 ? ? ? ParE_toxin ? Phage_integrase HTH_17 CbiA ? ? ?

NZ_PGGV01000038.1:142871-163156
Glycomyces xiaoerkulensis

? RibD_C ? DUF6247 GATase Chorismate_bind Anth_synt_I_N tRNA-synt_1 Anticodon_1 DUF5915 ParE_toxin HTH_OrfB_IS605 OrfB_IS605 Cas12f1-like_TNB DUF4878 PDDEXK_1 UvrD_C UvrD_C UvrD-helicase DrrA1-3_C ApbA_C ABC_tran PhdYeFM_antitox ? PDDEXK_1 UvrD_C UvrD-helicase Y1_Tnp

NZ_CAWNWM010000004.1:343081-354680
Acaryochloris thomasi RCC1774

DUF1508 DUF1508 Sacchrp_dh_C Sacchrp_dh_NADP ? HNH ABC-ATPase DUF4276 ParE_toxin ? ARG ? RecA_C RecA UvrD_C_2 AAA_30 Sigma70_r4_2 Sigma70_r3 Sigma70_r2 ? Bac_DNA_binding

NZ_QWDC01000001.1:1502127-1518362
Mucilaginibacter conchicola

Peripla_BP_6 Methyltr_RsmB-F MacB_PCD FtsX MacB_PCD FtsX MarC MFS_1 AA_permease_2 RelB-like ParE_toxin ? H_PPase AA_permease_2 ? ? ? Peptidase_S13

NZ_QQXK01000053.1:220-3632
Galactobacter valiniphilus

CSD ? ? ParE_toxin PhdYeFM_antitox RHH_1 FA_desaturase

NZ_CP033972.1:5949399-5962069
Gordonia insulae

Iso_dh Rv2993c-like_N FAA_hydrolase tRNA-synt_1c Anticodon_2 ? ? ParE_toxin PhdYeFM_antitox Gly_rich_2 Gp77 ? ? Cutinase Phage_holin_2 MFS_1 Abhydrolase_1

NZ_RQZF01000007.1:60008-82219
Schaalia canis

RAMPs RAMPs ? ? Pps Pps Pps PhdYeFM_antitox ABC_membrane ABC_tran CbiQ ABC_cobalt GDPD RAMPs ABC_membrane ABC_tran ABC_tran ABC_tran CRISPR_Cas2 ParE_toxin Cas_Cas1

NZ_AP017560.1:3856585-3867603
Pseudanabaena sp. ABRG5-3

DDE_Tnp_1 Bac_transf STAS Alg14 Glyco_tran_28_C DUF2214 HTH_5 SBF ? ParE_toxin ? ? XisH XisI DUF5763 AbiJ_NTD4 SLBB_2 Poly_export DUF2358 ?

NZ_PPPU01000016.1:206902-217472
Methylomonas koyamae

PEP-CTERM ? Exosortase_EpsH ? MqsA_antitoxin ParE_toxin ? HMA_2 HMA E1-E2_ATPase Hydrolase ? ? Phage_integrase Peptidase_M10 DsbD_2 ATPase-cat_bd ?

NZ_BJNA01000045.1:5970-7169
Microbacterium lacticum

HhH-GPD ? HGTP_anticodon tRNA-synt_2b DUF4870 Aldo_ket_red Acetylase ParE_toxin APH

NZ_BKAU01000001.1:2208899-2217799
Chitinophaga cymbidii

Zn_ribbon_recom Recombinase Resolvase ? SDH_protease DUF2158 ? ParE_toxin ? ? ? ? ? ? HTH_3 DNA_methylase DNA_methylase Vsr

NZ_VOSL01000140.1:18594-38937
Lujinxingia vulgaris

? RecC_C Exonuc_V_gamma PDDEXK_1 UvrD_C UvrD_C UvrD-helicase PhdYeFM_antitox ParE_toxin ? ? HTH_3 HigB-like_toxin HTH_3 Bestrophin-like Trypan_PARP RNase_H_2 UvrD_C_2 SH3_13 AAA_19 CvfB_WH S1_2 S1_2

NZ_WNKF01000004.1:173956-186725
Cyclobacterium roseum

MaoC_dehydratas ? Phage_int_SAM_1 Phage_integrase DHquinase_II Aminotran_5 MarC ? ParE_toxin ? LON_substr_bdg TPR_BSK1_C SH3_3 Cytochrome_CBB3 Pyrid_ox_like RelB-like Arm-DNA-bind_5 Phage_int_SAM_5 Phage_integrase

NZ_JAAKZI010000001.1:268746-280700
Arthrobacter silviterrae

DUF6278 ? HTH_3 PBP_N EamA EamA Cupin_2 HTH_3 ParE_toxin PhdYeFM_antitox ? ? ? ? GntR Aminotran_1_2 EamA EamA DUF2975

NZ_JAODLR010000018.1:89515-98731
Crystallibacter permensis

? P-II DUF1538 DUF1538 ? Abhydrolase_1 DUF7800 PhoD UPF0158 ParE_toxin Pyridox_ox_2 Pyridox_ox_2 DUF6176 BenE BenE PhdYeFM_antitox

NZ_CAKMSC010000019.1:1212-17634
Rothia nasimurium

ABC_tran SUFBD_core SufBD_N FeS_assembly_P Pyr_redox_dim Pyr_redox_2 HKD ParE_toxin PhdYeFM_antitox MerR_1 TipAS DUF6508 Mrr_N Mrr_cat PhdYeFM_antitox SUFBD_core ? Peptidase_M78 ?

NZ_JABVED010000002.1:58049-72019
Actinokineospora xionganensis

? TsaD TsaD ParE_toxin Hydrolase TsaE Ala_racemase_N Ala_racemase_C YjeF_N Carb_kinase GATase_6 SIS SIS Acetylase AMP-binding_C AMP-binding ACCA Carboxyl_trans Abhydrolase_6 Acetylase PhdYeFM_antitox

NZ_BMIK01000013.1:121744-138137
Parapedobacter defluvii

UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C Wzz Poly_export SLBB SLBB SLBB_2 SLBB_2 SLBB SLBB HTH_DeoR AAA_14 DUF4143 Rrf2 ParE_toxin RelB-like DUF748 Ldh_2 Peptidase_M1 HupE_UreJ_2 DUF6702 SLFN_AlbA_2 HATPase_c_4 HTH_24 23S_rRNA_IVP

NZ_CP071974.1:2129353-2149578
Trueperella pecoris

PhdYeFM_antitox ? Lipocalin_2 TetR_N TrkA_C TrkA_N 2-Hacid_dh_C CRISPR_Cas2 Cas_Cas1 Cas9_C Cas9_Topo Cas9_b_hairpin RuvC_III HNH Cas9_a ParE_toxin TrkH ? rve rve_3 tRNA_U5-meth_tr tRNA_U5-meth_tr HrpB_C Helicase_C DEAD

NZ_AP025343.1:1195992-1208273
Paenibacillus azoreducens

DUF899 EVE EVE ? Germane Cu_amine_oxidN1 AMIN Amidase_3 Acetylase ParE_toxin GNAT_acetyltran Methyltransf_25 HxlR KIF21A_4th Sigma70_r4 Sigma70_r2 ? Spc24 BRI3BP

NZ_JADDUE010000007.1:204127-215001
Saccharopolyspora galaxeae

LacI Peripla_BP_3 zf-HC3 HTH_31 DUF5753 DUF397 ? DLH PhdYeFM_antitox ParE_toxin Fapy_DNA_glyco H2TH zf-FPG_IleRS p450 DUF6545 ? ? Abhydrolase_1 3HCDH_N 3HCDH

NZ_CP054142.1:1452816-1469100
Treponema parvum

DUF6290 DUF2281 ? PIN ? DUF4474 ? ParE_toxin RHS AMP-binding AMP-binding_C_3 MGDG_synth Glyco_tran_28_C HigB-like_toxin AMP-binding AMP-binding_C_3 ? ThiI THUMP ThiI_fer

NZ_CP083582.1:4649621-4668512
Kovacikia minuta CCNUW1

CRISPR_Cas2 CARF_Card1 Cas10-Cmr2_palm2 Cmr2_N Cas_Cmr3 Cas_Cas02710 ParE_toxin DUF2281 ParE_toxin Cas10-Cmr2_palm2 Cmr2_N RAMPs ? ? Cas_Cmr3 Cas_Cas1 ?

NZ_JAJKFK010000005.1:185036-199329
Amycolatopsis sp. GM8

p450 p450 ? ? ? ? Phage_int_SAM_4 PaaA2 ParE_toxin HAD_2 GntR HTH_17 Sigma70_r2 Sigma70_r4_2 HTH_17 AAA_25 HTH_17 Phage_int_SAM_3 Phage_integrase Phage_integrase

NZ_JAQMSG010000107.1:677-13322
Nodularia spumigena CS-591/07A

TPRs Glyco_transf_41 Glyco_transf_41 HTH_DNAJC9 ParE_toxin ? Psb28 Glyco_trans_1_4 TylF Methyltransf_24 Glycos_transf_1 Glyco_transf_4 N_methyl Pilin_GH MoCF_biosynth ? Glyco_transf_41 Glyco_transf_41

NZ_CP112874.1:4468736-4483039
Pseudanabaena galeata CCNP1313

? His-Kinase His-Kinase PAS HAMP Uma2 Yae1_N ? PhdYeFM_antitox BCA_ABC_TP_C ABC_tran ParE_toxin ? DUF3326 MFS_1 MFS_1 Pps Pps Pps Pps Pps TerC FAD_binding_3 Glyoxalase

NZ_JAAHGX010000467.1:107-1088
Okeania sp. SIO2G5

ParE_toxin Uma2

NZ_JAMJFP010000074.1:6644-16150
Methylobacter sp.

KleE ? DUF6792 GFA ? TMA7 ParE_toxin RPA HTH_3 KORA T2SSE TrbC VirB3 ?

NZ_WHTD01000148.1:11679-20962
Actinophytocola sp.

CorC_HlyC CBS CBS CNNM CNNM ? SecDF_P1_head PCMT ParE_toxin ? ? ? DUF6879 MDMPI_N Transglut_core3 Transglut_core3

NZ_JAQSCS010000017.1:31571-50454
Methylobacter sp.

DUF4912 GH94_b-supersand AAA_15 AAA_15 OLD-like_TOPRIM Unstab_antitox ? RelB-like ParE_toxin Y1_Tnp Glyco_hydro_30C Glyco_hydro_30 ISOA1-3_C Alpha-amylase CBM_48 ChlI AAA_25 Zn_ribbon_LapB Zn_protease CorA Peptidase_M28 BE_C Glyco_hydro_57

NZ_CP129884.1:439764-453396
Methylomonas montana

Cas_Cas5d CRISPR_assoc MazE_antitoxin PIN HTH_37 DUF6290 ParE_toxin Cas_Cas1 Cas_Cas1 Cas_Cas2CT1978 Fer4_21 FeS tRNA_bind Anticodon_3 tRNA-synt_1g ? ? SPOR Ldt_C YkuD LysM CRISPR_assoc

NZ_CP136520.1:2571986-2585574
Nodosilinea sp. E11

Ferritin FUR ? Aldedh ThiC_Rad_SAM DUF4383 SSF ParE_toxin ? ? ACT_7 ACT_3 ? AhpC-TSA PTP-NADK HTH_31 LRR_14 LRRs AP-GTPase COR-A COR-B TIR

NZ_JAYGHW010000011.1:22007-38933
Spirulina sp. CCNP1310

DHBP_synthase GTP_cyclohydro2 EVE ? ? ? ? ParE_toxin ? ? PemK_toxin ? RNaseH-CATRA+Caspase TPRs TPRs TPRs TPRs TPRs TPRs PPC Trypsin MutS_V MutS_III MutS_II MutS_I Fer4_5 cNMPBD

NZ_JAYFUM010000014.1:143393-155569
Arcicella rigui

PIN Arginosuc_synth Arginosuc_syn_C DUF2281 PIN Semialdhyde_dh Semialdhyde_dhC_1 ? ? ParE_toxin Aminotran_3 Peptidase_S9 DPPIV_N ? LuxE MoaE DHFR_1 Formyl_trans_N ACT

NZ_JAYGIN010000004.1:91194-103788
Arcicella sp. LKC2W

ACP_syn_III ACP_syn_III_C HigB_toxin HTH_3 Fe-S_biosyn UbiA GSHPx Arg_tRNA_synt_N tRNA-synt_1d DALR_1 ParE_toxin ? ? ? Sua5_yciO_yrdC tRNA_NucTran2_2 HD PolyA_pol_RNAbd PolyA_pol AP_endonuc_2 TAT_signal adh_short_C2